KERALA SSLC | PHYSICS

SOUND WAVES

PYQ • 2021 - 2026

1. 2026 **3 MARKS**

Observe the illustration of a wave.

(a) Which type of mechanical wave is this ?

(b) What do the letters C and R indicate ?

(c) If the distance between two consecutive 'C' of the wave is 0.01 m, which is travelling with a speed of 350 m/s, will it be audible to humans ? Justify your answer by mathematical calculation.

2. 2026 **2 MARKS**

A child standing between two large buildings claps and hears the echo from the first building in 0.2 seconds and the echo from the second building in 0.3 seconds. (Speed of sound in air = 340 m/s)

(a) What is the condition to produce an echo ?

(b) Calculate the distance between the two buildings.

3. 2026 **1 MARK**

Statement: If the stem of a tuning fork is placed on a table, the sound heard increases.

Reason: If the frequency of the forcing object and the natural frequency of the forced object are not equal, the objects are said to be in resonance. The frequency of the object undergoing resonance will increase.

(a) Both the Statement and Reason are correct.

(b) Both the Statement and Reason are incorrect.

(c) The Statement is correct but the Reason is incorrect.

(d) The Statement is wrong but the Reason is correct.